Python Programming Internship - Week 1

Project Title

Conditional Statements and Loops in Python

Objective

The objective of this project is to learn Python control flow using conditional statements and loops.

Tasks Completed

1. Number Classification

This program checks whether a number is:

* Even or Odd
* Positive, Negative, or Zero

File: src/number_checks.py

2. Mathematical Loops and Patterns

This program:

* Generates a multiplication table
* Calculates the sum of the first N natural numbers
* Prints a right-angled triangle pattern using asterisks

File: src/loop_operations.py

Concepts Used

* if, elif, else
* for loop
* User input
* Arithmetic operators
* Pattern printing

How to Run

Open the terminal and run:

python src/number_checks.py

python src/loop_operations.py

Learning Outcome

I learned how to use conditional statements and loops in Python to solve basic logical problems and improve problem-solving skills.

src new folder

.py | loop_operations.py | README.md | .gitignore

C: > Users > abhis > Desktop > Python Programming Internship - Week 1 > .gitignore

```
__pycache__/
*.pyc
.vscode/

```

File Edit Selection View Go Run ...

EXPLORER

SRC NEW FOLDER

- loop_operations.py
- number_checks.py

number_checks.py | loop_operations.py | README.md

number_checks.py > ...

```python
# Number Classification Program

num = int(input("Enter an integer: "))

# Even or Odd
if num % 2 == 0:
    print("The number is Even")
else:
    print("The number is Odd")


# Positive, Negative or Zero
if num > 0:
    print("The number is Positive")
elif num < 0:
    print("The number is Negative")
else:
    print("The number is Zero")
```

TIMELINE

PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS

```
Enter an integer: 10
The number is Even
The number is Positive
PS C:\Users\abhis\Desktop\src new folder>
```

File Edit Selection View Go Run ...

src new folder

number_checks.py | loop_operations.py | README.md

SRC NEW FOLDER

- loop_operations.py
- number_checks.py

loop_operations.py > ...

```python
num = int(input("Enter a number: "))

print("\nMultiplication Table")

for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")

# Sum of first N natural numbers

n = int(input("\nEnter N: "))

total = 0

for i in range(1, n + 1):
    total += i

print("Sum =", total)

# Star Pattern

rows = int(input("\nEnter number of rows: "))

print()

for i in range(1, rows + 1):
    print("*" * i)
```

PROBLEMS DEBUG CONSOLE OUTPUT TIMELINE TERMINAL PORTS

```
Multiplication Table
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```